

AVINASH PATIL

Java Backend Developer | Spring Boot | Microservices | 3.5+ Years

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PROFILE SUMMARY

Results-driven **Java Backend Developer** with **3.5+ years** of professional experience designing, developing, and deploying scalable microservices and event-driven systems across **BFSI** and **IoT** domains. Proficient in **ReactJs**, **Java 8/11/17**, **Spring Boot**, **Spring Security**, **Spring Data JPA**, **Hibernate**, and **RESTful API** development, with hands-on expertise in **Apache Kafka**, **PostgreSQL**, **MySQL**, **Docker**, **Kubernetes**, and **AWS** (EC2, S3, Lambda, CloudWatch). Architected multi-tenant platforms processing **100,000+ daily events** at **99.9% uptime**, optimized database performance by **65%**, and delivered **88%+ test coverage** through JUnit 5 and Mockito. Skilled in CI/CD pipelines via Jenkins and GitLab, Agile/Scrum delivery, performance optimization, and security implementation using **JWT**, **OAuth2**, and **RBAC**. Passionate about clean code, SOLID principles, and engineering high-availability backend systems that drive measurable business impact.

TECHNICAL SKILLS

Programming Languages: Java 8, Java 11, Java 17, Java 21, SQL, PL/SQL, JavaScript (ES6)
Backend Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Spring Cloud, Spring Batch, Spring Cloud Gateway, Hibernate ORM, Resilience4j, Struts
Frontend Technologies: ReactJs, JSP, Servlet, HTML5, CSS3, Bootstrap, AJAX, jQuery
Databases: PostgreSQL, TimescaleDB, MySQL, Oracle
Caching: Redis (Cache-Aside Pattern)
Messaging & Event Streaming: Apache Kafka, MQTT
Cloud (AWS): EC2, S3, RDS, Lambda, ECS, IAM, CloudWatch, EBS, API Gateway
DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, SonarQube, Shell Scripting
Testing: JUnit 5, Mockito, Integration Testing
Security: Spring Security, JWT, OAuth2, RBAC, AES-256, PCI-DSS Compliance
API & Documentation: REST, OpenAPI 3.0, Swagger UI, Postman, Feign Client
Build Tools: Maven
Version Control: Git, GitHub, GitLab
Monitoring & Logging: SLF4J, Log4j, Logback, AWS CloudWatch
Design & Architecture: Microservices, Event-Driven Architecture, Multi-Tenant Architecture, SOLID Principles, OOP, Design Patterns (Singleton, Factory, Observer, Strategy, Builder), Distributed Systems
Servers: Apache Tomcat, Oracle WebLogic
IDE & Tools: IntelliJ IDEA, Eclipse, VS Code, Postman, DBeaver, pgAdmin
Operating Systems: Linux, Unix, Windows

PROFESSIONAL EXPERIENCE

Pratiti Technologies Pvt. Ltd.

Feb 2026 – Present

Software Engineer – Java Backend

Pune, Maharashtra

- **Energy Digital Twin Platform – IoT / Industrial Automation**
- Architected and deployed **7 production-grade event-driven microservices** in Java 17 processing **100,000+ daily IoT telemetry events** with **sub-50ms P99 latency** and **99.9% uptime SLA** using Spring Boot, ExecutorService thread pool optimization, and async processing patterns.
- Designed and implemented real-time telemetry ingestion pipeline (**MQTT + Apache Kafka**) with message deduplication, idempotency checks, and exactly-once semantics, reducing **duplicate events by 99.2%** across **10,000+ connected IoT devices**.
- Optimized **PostgreSQL + TimescaleDB** time-series schema handling **100M+ daily inserts**, achieving **sub-200ms query latency** through strategic B-tree indexing, timestamp-based partitioning, and table inheritance strategies.
- Implemented granular **Role-Based Access Control (RBAC)** system with 8 role types, JWT token validation with refresh mechanisms, and dynamic permission evaluation, securing **50+ REST API endpoints** across multi-tenant infrastructure serving 500+ organizations.
- Designed comprehensive **OpenAPI 3.0** documentation with Swagger UI for 25+ endpoints, improving third-party integration efficiency by **35%** and reducing developer onboarding time.
- Containerized all microservices using **Docker** and orchestrated with **Kubernetes** (HPA, rolling deployments), enabling **zero-downtime deployments** and horizontal auto-scaling, reducing deployment time by **50%**.
- Conducted **code reviews** for 8 junior engineers, mentoring on microservices design patterns, SOLID principles, REST API best practices, and database optimization strategies.
- Collaborated in **Agile/Scrum** sprints – sprint planning, daily standups, code reviews, retrospectives – maintaining consistent sprint velocity across all delivery cycles.

Diensten Tech Ltd.

Java Developer

Feb 2025 – Feb 2026

Pune, Maharashtra

- **Core Banking Solutions – TCS BaNCS – Banking / Transaction Processing / Regulatory Compliance**
- Developed and maintained **12 production microservices** for TCS BaNCS core banking platform processing **50,000+ daily transactions** worth **USD 2M+** across NEFT, RTGS, IMPS rails, serving **500+ branches** and **100,000+ active customers** with **99.95% uptime SLA** and **sub-100ms latency**.
- Designed and developed **18+ RESTful APIs** with full OpenAPI 3.0 documentation, achieving **92% test coverage** using JUnit 5, Mockito, and AssertJ, identifying and eliminating 12 pre-production defects.
- Implemented **circuit breaker pattern** using Resilience4j with configurable failure thresholds and retry logic, reducing **cascading failure risk by 95%** and enabling graceful service degradation.
- Delivered **PCI-DSS and RBI-compliant** transaction processing with AES-256 encryption, HSM-based key management, and complete audit logging of all sensitive financial operations, achieving **zero security incidents**.
- Achieved **88% code coverage** across **45,000+ lines** of production code via JUnit 5 and Mockito, identifying **12 critical pre-production bugs** before production release.
- Executed **3 complete UAT cycles** for core banking features (deposits, withdrawals, fund transfers), coordinating with QA and business stakeholders, resulting in **zero post-release defects**.
- Automated **CI/CD pipeline** using Jenkins + GitLab CI with SonarQube integration, reducing **deployment time by 40%** and enforcing consistent code quality gates.

Maxgen Technologies Pvt. Ltd.

Jr. Java Developer

Sep 2022 – Jan 2025

Pune, Maharashtra

- **Core Banking Solutions – TCS BaNCS – Account Management / Customer Lifecycle**
- Developed **8 production backend modules** for TCS BaNCS platform (customer onboarding, KYC, account management, deposits, withdrawals, loans) using Java 8, Spring Boot 2.x, Hibernate ORM, and Oracle, supporting **100,000+ active banking customers**.
- Resolved **45+ production defects** through systematic root cause analysis, defensive coding improvements, and enhanced exception handling, reducing **bug recurrence rate by 65%**.
- Optimized slow-running Oracle database queries via **EXPLAIN PLAN** analysis, strategic B-tree indexing, and SQL restructuring, improving **query execution speed by 65%** and reducing backend response times.
- Implemented **RESTful APIs** for authentication, authorization, and third-party integrations (KYC providers, payment gateways), standardizing API design across a 6-member engineering team.
- Improved **unit and integration test coverage** by writing comprehensive JUnit test suites, reducing regression defects by **40%** and improving overall code quality.
- Participated actively in **Agile/Scrum** ceremonies (sprint planning, standups, reviews, retrospectives), maintaining **95% sprint velocity** consistency across **12+ consecutive sprints**.
- Mentored 2 junior developers on Spring Boot fundamentals, debugging techniques, and testing best practices, reducing their ramp-up time by **30%**.

KEY ACHIEVEMENTS

- Architected scalable **multi-tenant IoT platform** processing **100,000+ daily telemetry events** with **99.9% uptime**, reducing time-to-insight from 5 minutes to **sub-500ms** through event-driven architecture and real-time streaming.
- Built resilient microservices with **circuit breaker patterns**, retry mechanisms, and graceful degradation using Resilience4j, **preventing cascading failures** and reducing production incidents by **65%**.
- Optimized application performance through **Redis caching** (cache-aside pattern), database query optimization using indexing and execution plan analysis, and async processing, **reducing latency by 45–68%**.
- Domain expertise in **core banking systems** (TCS BaNCS), **IoT platforms**, event-driven architecture, multi-tenant systems, and distributed systems across BFSI and Industrial Automation sectors.
- Achieved **88–92% code coverage** through disciplined testing practices (JUnit 5, Mockito, AssertJ, TestContainers), preventing pre-production bugs and maintaining **zero post-release defects** across multiple UAT cycles.

EDUCATION

R. C. Patel Institute of Technology, Shirpur

Bachelor of Technology in Computer Engineering

2018 – 2022

CGPA: 8.57 / 10